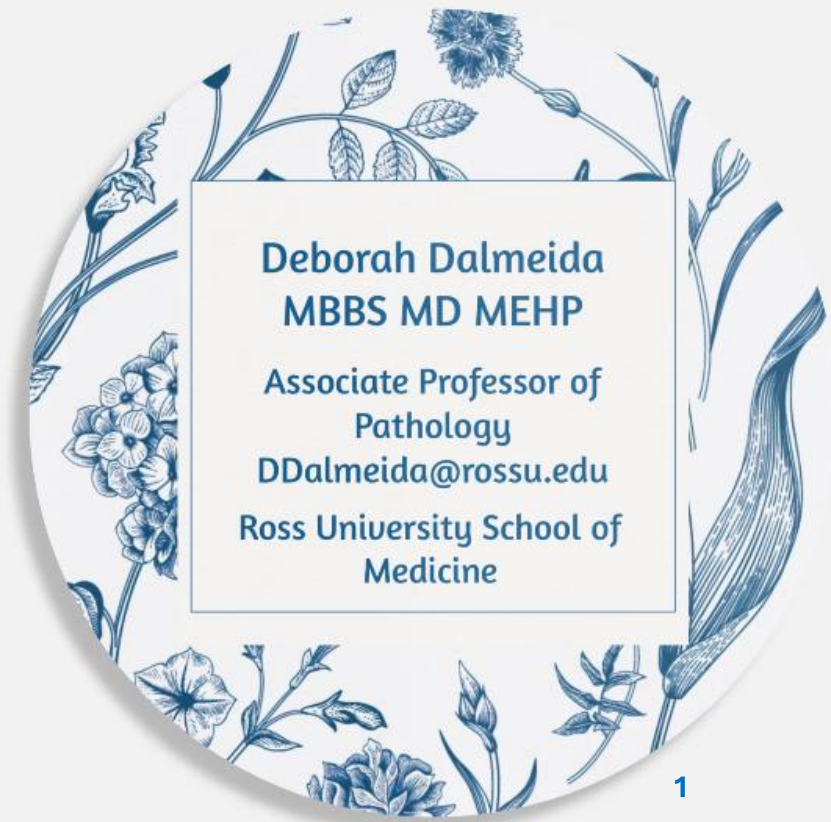


1. Disorders of the hypothalamus and pituitary gland -1





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Retrieval activity



Anterior pituitary hormones

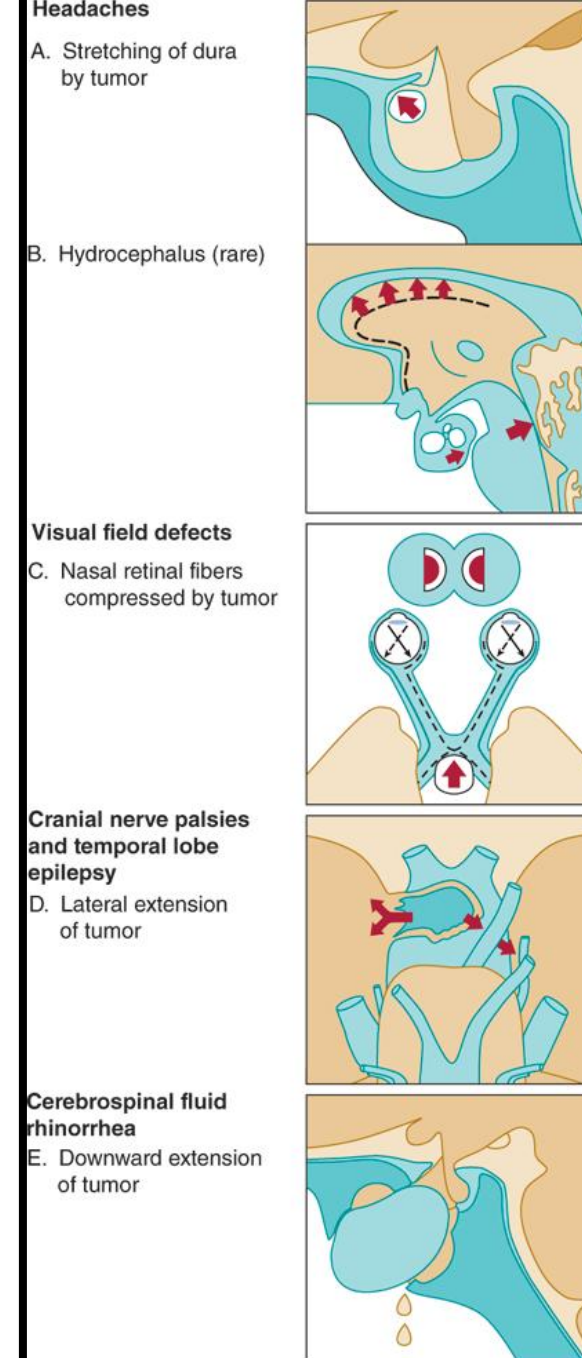
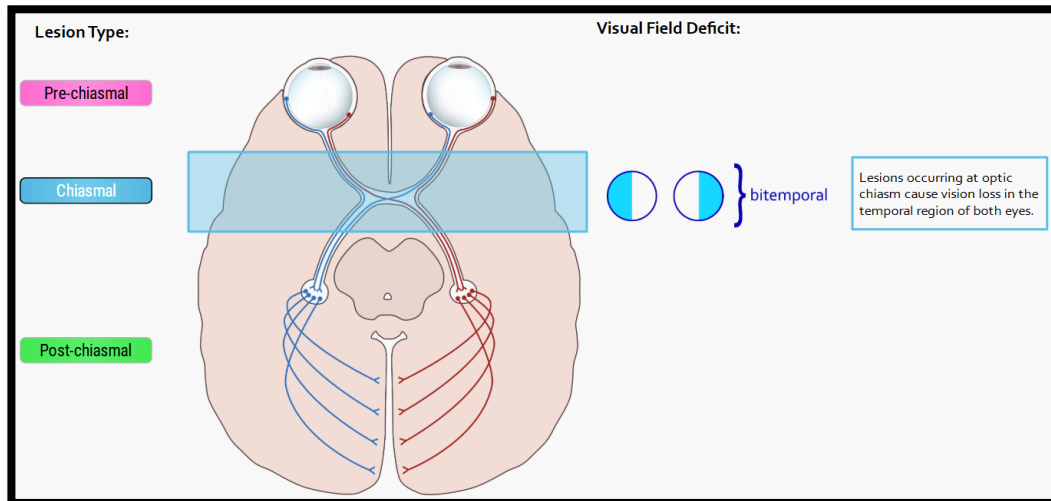
	ACTH	GH	Prolactin	TSH	LH	FSH
Peptides	Derived from POMC precursor	Single-chain polypeptide	Single-chain polypeptide	α : $\alpha 1$	α : $\alpha 1$	α : $\alpha 1$
				β : TSH- β	β : LH- β	β : FSH- β
Receptor	ACTH-receptor (melanocortin-2-receptor)	GH receptor	Prolactin receptor	TSH receptor	LH receptor	FSH receptor
Source	Corticotropes (pituitary)	Somatotropes (pituitary)	Lactotropes (pituitary)	Thyrotropes (pituitary)	Gonadotropes (pituitary)	Gonadotropes (pituitary)
Hypothalamic releasing hormone	CRH, AVP	GHRH (ghrelin)	TRH	TRH	GnRH	GnRH
Hypothalamic inhibiting factors		Somatostatin	Dopamine	Somatostatin, dopamine		
Target	Adrenal gland	Liver (production of IGF-1), peripheral tissue	Mammary gland	Thyroid gland	Ovary (theca cell, granulosa cell, luteal cell)/testis (Leydig cell)	Ovary (granulosa cell)/testis (Sertoli cell)
Function	Stimulating cortisol and adrenal androgen release	Stimulating growth (direct and indirect effect via IGF-1)	Stimulating lactation	Stimulating thyroid hormone release	Stimulating estrogen/testosterone production	Regulating granulosa and Sertoli cell function

Pituitary adenomas

- benign tumor of epithelial cell origin
- **Clinical significance**
 - Enclosed space within which the pituitary is situated has a limited capacity to accommodate an expanding mass
 - May arise from cells secreting hormones → hormone overproduction
 - Destruction of the normal pituitary by the expanding adenoma → hormone deficiency
- Seen in adults at 35-60 years of age
- Microadenomas <1cm in diameter
- Macroadenomas >1cm in diameter
- Majority- clinically silent
- Functional – PRL+ GH most common
- Clinical symptoms: Endocrine symptoms (*based on cell type from which the tumor originated*) + local mass effect

Pituitary adenomas

- Expanding intracranial mass → headache
- Visual field defects – **bitemporal hemianopia**
 - Tumor compresses the crossing fibers of the optic tract, which lie directly above the pituitary gland and innervate the part of the retina responsible for temporal vision.



Pituitary adenomas

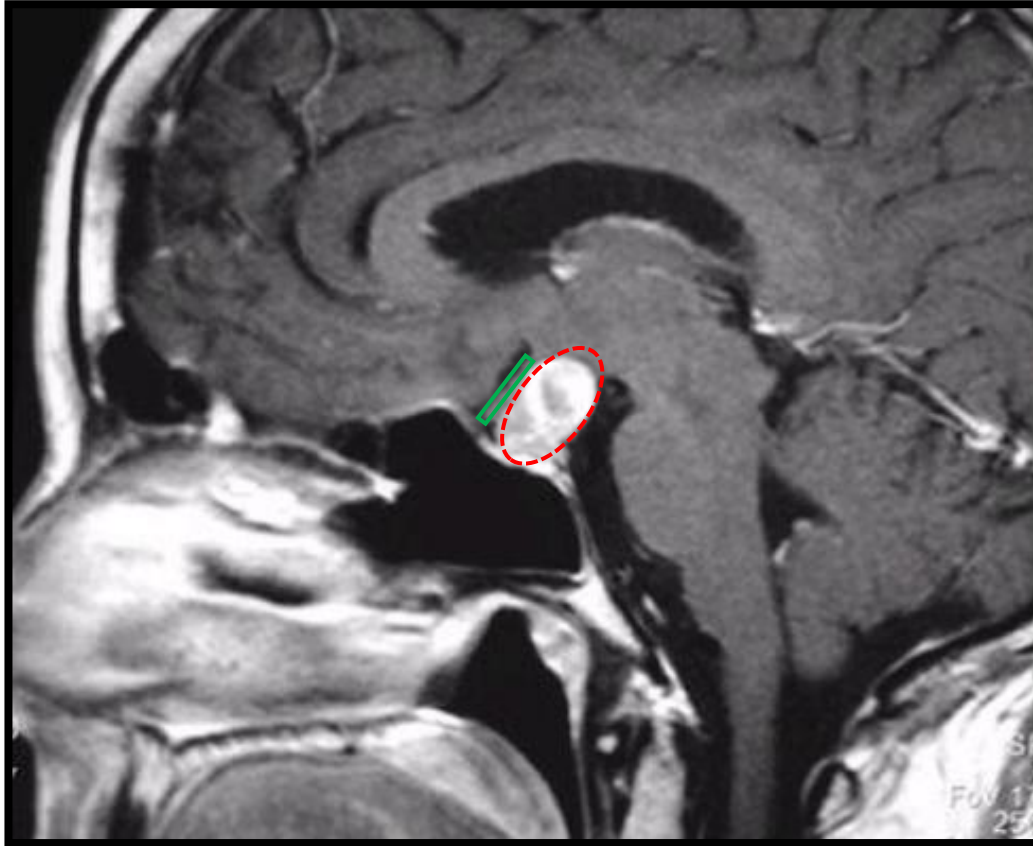
- **Pathogenesis:**

- Somatic *GNAS* mutations → nullifies GTPase activity of G_{α} → constitutive activation of G_{α} → persistent generation of cAMP → unchecked cellular proliferation
- Germline loss-of-function mutation in genes such as *MEN1*

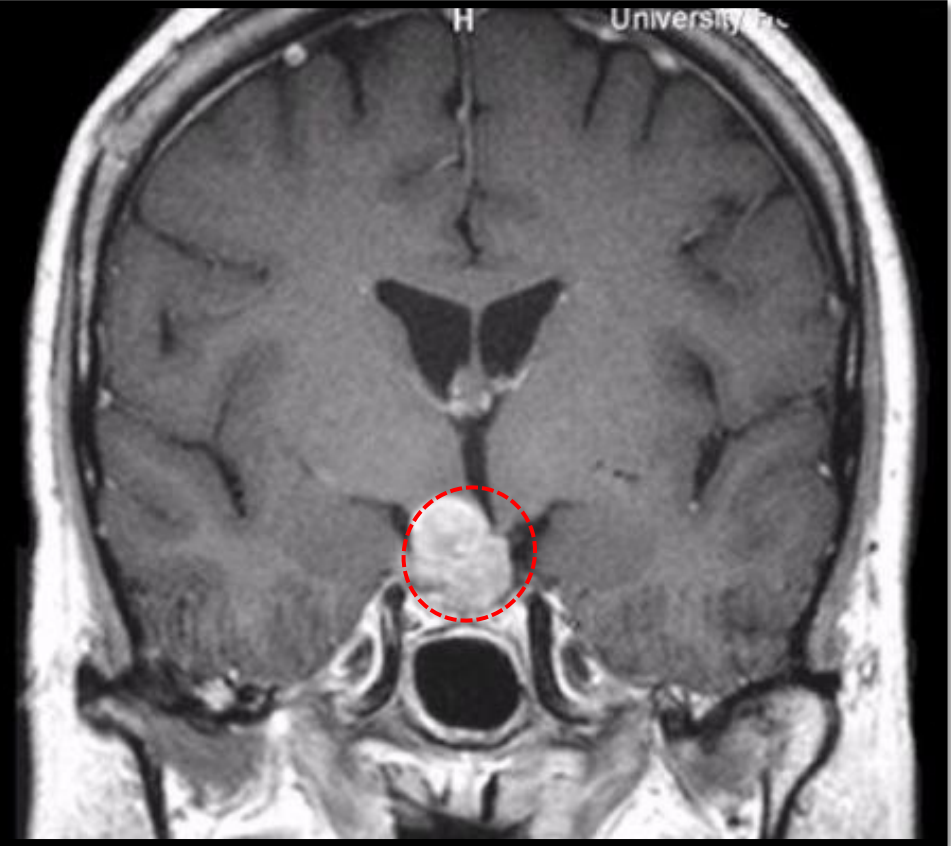
- **MEN1 syndrome**

- Autosomal dominant
- germline mutations in the *MEN1* tumor suppressor gene, which encodes a protein called menin.
- Components include: **3Ps**
 - Pituitary adenomas
 - Parathyroid adenomas
 - Pancreatic endocrine tumors – Zollinger Ellison syndrome, insulinomas, VIPomas, glucagonomas

Pituitary adenomas



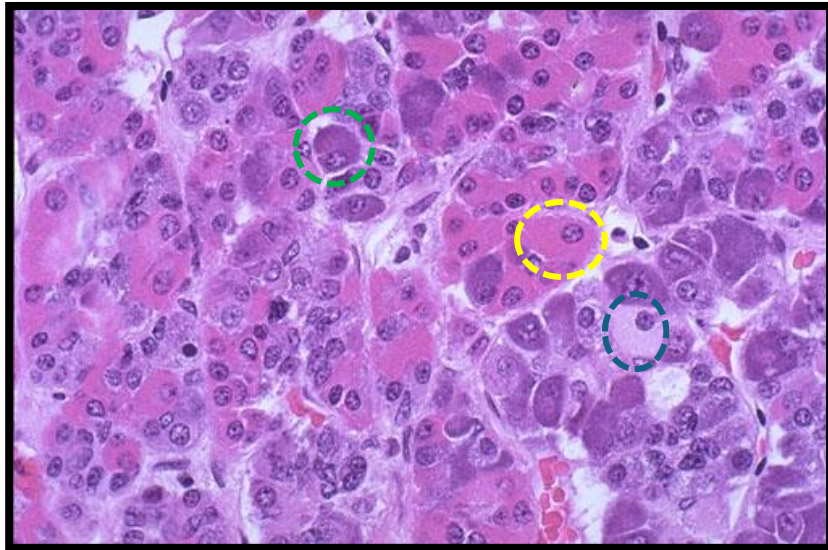
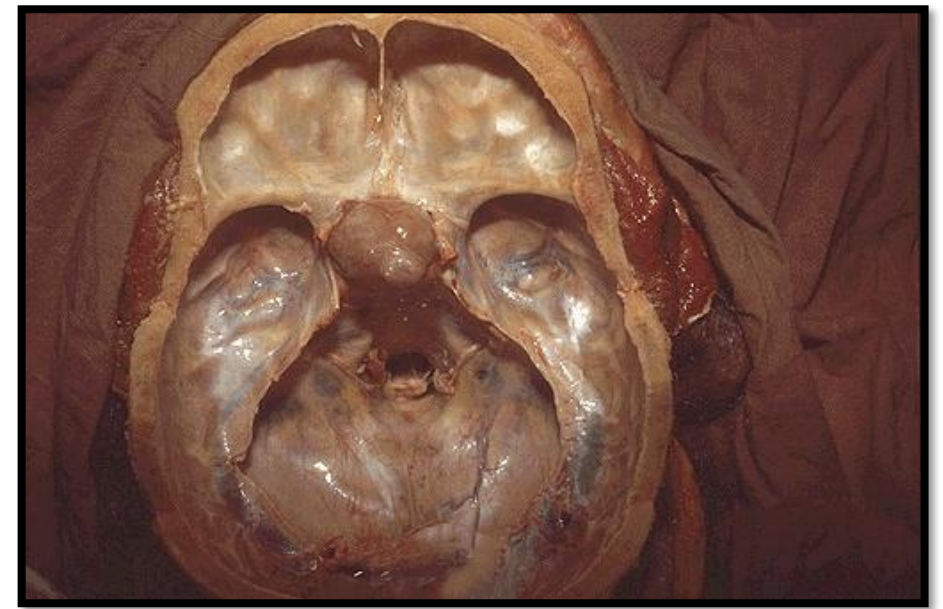
T1-weighted with gadolinium sagittal MRI. Note displacement of the optic chiasm (green) and enhancing mass (red)



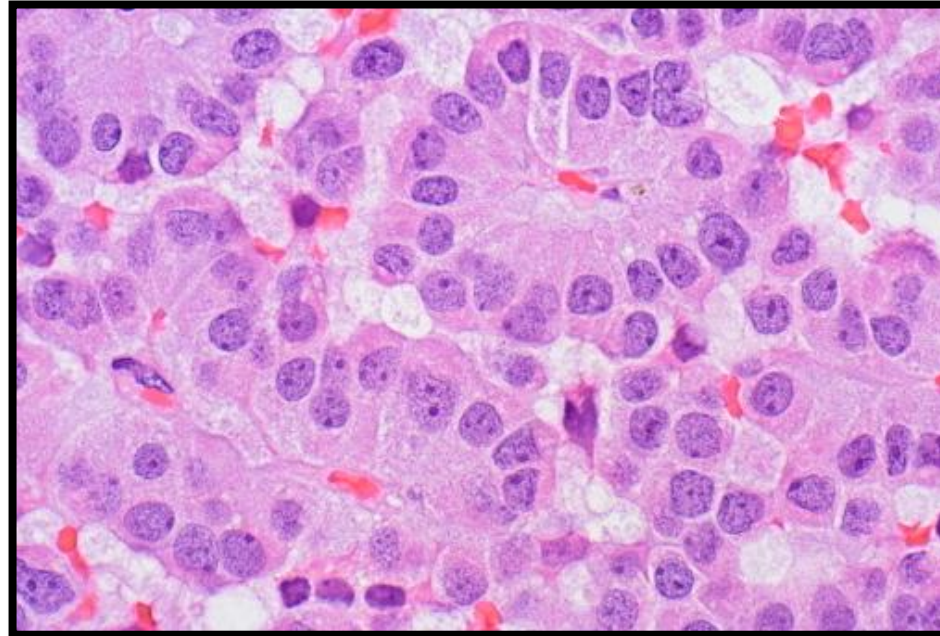
T1-weighted with gadolinium coronal MRI. Note the large enhancing mass (red) in the region of the sella that is growing up into and displacing the optic chiasm and hypothalamus.

Pituitary adenomas

- **Gross morphology** : soft, well-circumscribed
- **Microscopy**: monomorphic, polygonal cells arrayed in sheets or cords with sparse connective tissue



Histology of the anterior pituitary



High power microscopic view of a pituitary adenoma. Small round cells with small round nuclei and pink to blue cytoplasm. The cells may be arranged in nests or cords and endocrine tumors also have prominent vascularity.

Lactotroph adenoma

- most common type of hyperfunctioning pituitary adenoma
- Secrete prolactin
- Serum prolactin concentrations tend to correlate with the size of the adenoma.

Clinical presentation:

- Galactorrhea

▪ Women

- PRL inhibits both the normal pulsatile secretion of LH and FSH and the midcycle LH surge → oligomenorrhea with anovulation
- Decreased FSH → decreased estrogen; Decreased LH → decreased progesterone → both contribute to → secondary amenorrhea

- Estrogen deficiency → osteoporosis
- **Men**
- PRL excess → decreased testosterone synthesis and decreased spermatogenesis → decreased libido, impotence, and infertility.

Somatotroph adenoma

- Second most common type of hyperfunctioning pituitary adenoma
- secretes growth hormone
- Elevated GH → increased hepatic secretion of IGF-1

Clinical presentation:

- Children – prior to epiphyseal closure at growth plates → increased linear bone growth → gigantism
- Adults - acromegaly

Acromegaly

- Increased lateral bone growth - large, thickened, and bulky, with blunt, sausage-like fingers and toes
- Increases in ring, glove, and shoe sizes
- Coarsening of facial features, frontal bossing (increased hat size)
- Prominent supraorbital ridges; enlargement of the nose; downward and forward growth of the mandible → prognathism
- Widely spaced teeth, macroglossia
- Thickened skin with increased oiliness and sweating - bulky, sweaty handshake
- Acanthosis nigricans, skin tags

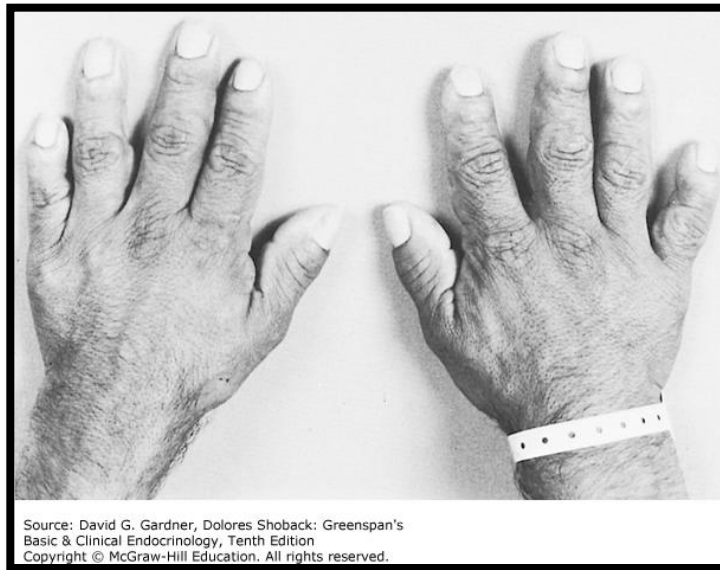
Somatotroph adenoma

Laboratory findings:

- elevations in IGF-1 and GH
- Elevated insulin
- Elevated glucose
- Failure to suppress GH production in response to an oral load of glucose



Acromegalic facies (photo). Case study, Radiopaedia.org (Accessed on 30 Jun 2026) <https://radiopaedia.org/cases/36249>



1977



1981



1983



1988

Gradual change in facial features in acromegaly (prior to diagnosis). Molitch ME. Clinical manifestations of acromegaly. Endocrinol Metab Clin North Am. 1992;21(3):597-614.

Somatotroph adenoma

Complications:

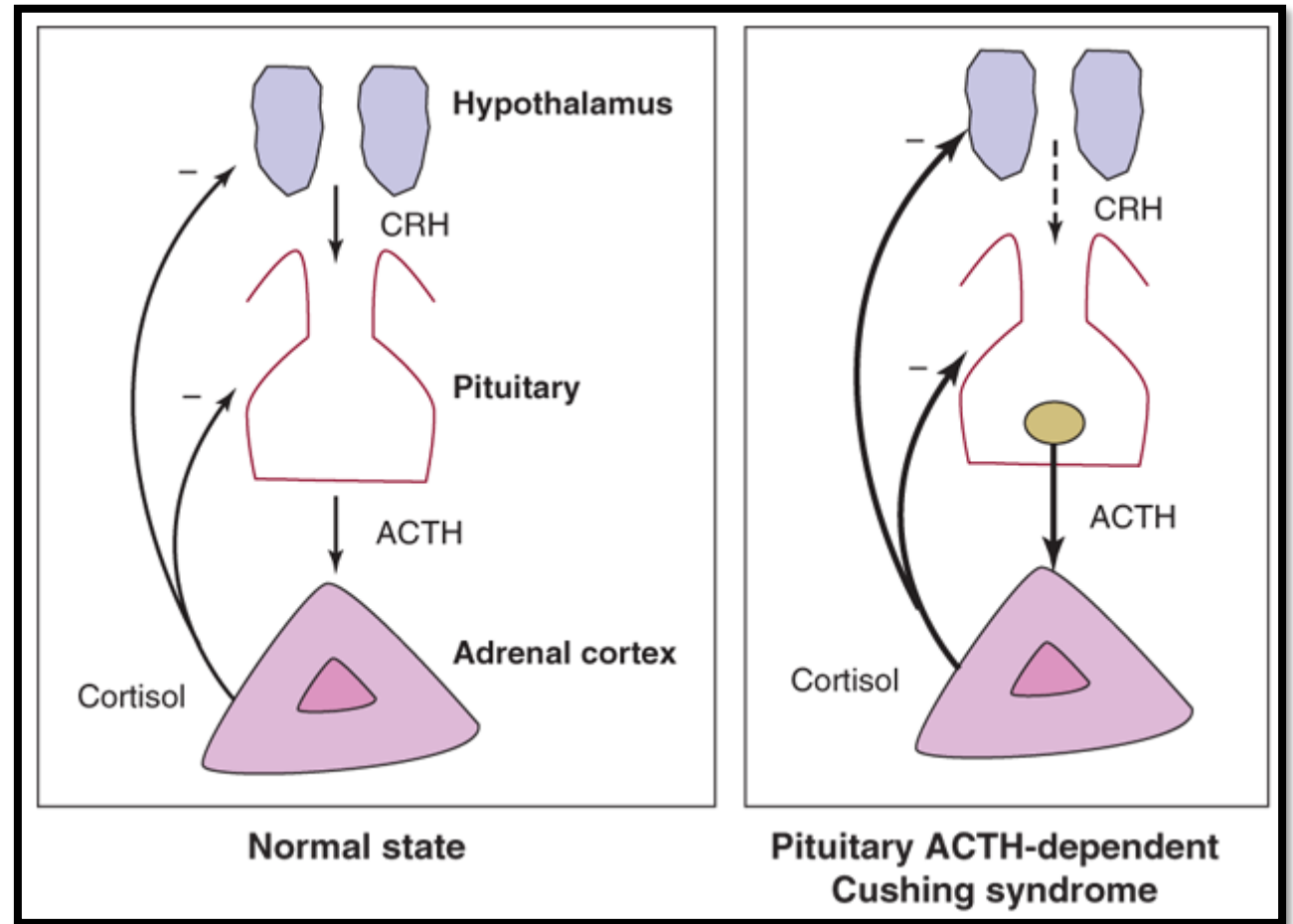
- GH-induced insulin resistance → hyperinsulinemia, glucose intolerance, **diabetes mellitus**, ketoacidosis
- Insulin increase → sodium retention → **hypertension**
- Increased risk of colonic polyps and tumors
- Visceral organomegaly with dysfunction- heart, kidneys, liver, thyroid
- **Dilated cardiomyopathy is the most common cause of death**

Corticotroph adenoma

- Secretes **ACTH**
- Cushing disease = hypercortisolism due to excessive production of ACTH by the pituitary
- Most common spontaneous cause of Cushing syndrome

**Other causes of Cushing syndrome will be discussed in disorders of the adrenal gland*

- Male to female ratio is 5:1



Corticotroph adenoma

1. Hypercortisolism → suppression of CRH and hypothalamic function → loss of hypothalamic control of ACTH secretion → **stressful stimuli** (hypoglycemia/ surgery) fail to elevate ACTH and cortisol secretion
2. Increased adrenal androgen secretion → peripheral conversion to T and DHT → androgen excess
 - Women - **hirsutism, acne, and amenorrhea**
 - Men - **decreased libido and impotence**

3. Systemic effects of cortisol excess

- **Obesity – central type** with sparing of extremities
- **Moon facies** with fat accumulation around the neck in the supraclavicular and dorsocervical fat pads
- Skin thinning → **red-purple abdominal striae**, easy bruisability
- Acne
- Delayed wound healing, frequent mucocutaneous fungal infections
- Hyperglycemia
- Hypertension: DBP > SBP
- Osteoporosis
- Proximal muscle weakness

Corticotroph adenoma



Central obesity in Image A



Multiple wide purplish striae.



Fat deposition in the face results in the classic finding of “moon facies.”



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Hypopituitarism

- diminished or absent secretion of one or more pituitary hormones.
- **Hypothalamic versus pituitary causes**
 - **Majority**- loss of 75% parenchyma of anterior pituitary
 - **Suspect hypothalamic origin** when there's evidence of anterior+ posterior pituitary dysfunction
- **Etiology: Nine I's**

1. **Invasive** - Large pituitary adenomas, craniopharyngioma, primary CNS tumors
2. **Infarction** – Sheehan syndrome, pituitary apoplexy
3. **Infiltrative** – Sarcoidosis, hemochromatosis, Langerhan histiocytosis
4. **Injury** - severe head trauma
5. **Immunologic** - Lymphocytic hypophysitis
6. **Iatrogenic** – surgical/ radiation therapy to the pituitary gland
7. **Infectious** (rare)
8. **Idiopathic**
9. **Isolated** – Kallmann syndrome, anorexia nervosa, marked obesity

Ischemic necrosis of the pituitary (Sheehan syndrome)

- Pregnancy → pituitary gland undergoes physiologic expansion (2x) but there's no increase in blood supply → relative hypoxia → [obstetric hemorrhage/ shock](#) → anterior pituitary lobe infarction
- Disseminated intravascular coagulation
- Sickle cell anemia
- Traumatic injury
- Elevated intracranial pressure
- Shock

What is the most likely histologic pattern of necrosis ?



Pituitary apoplexy

- Acute neurosurgical emergency
- Spontaneous hemorrhagic infarction of a pituitary adenoma + acute hypopituitarism

Clinical presentation:

- severe headache, visual impairment, ophthalmoplegias, meningismus, and an altered level of consciousness
- Hypotension, rapid mental deterioration, coma
- Death

Sarcoidosis

- Hypothalamus is the most frequently involved endocrine gland in sarcoidosis

Clinical presentation

- Diabetes insipidus
- Pituitary stalk infiltration → galactorrhea

Kallmann syndrome

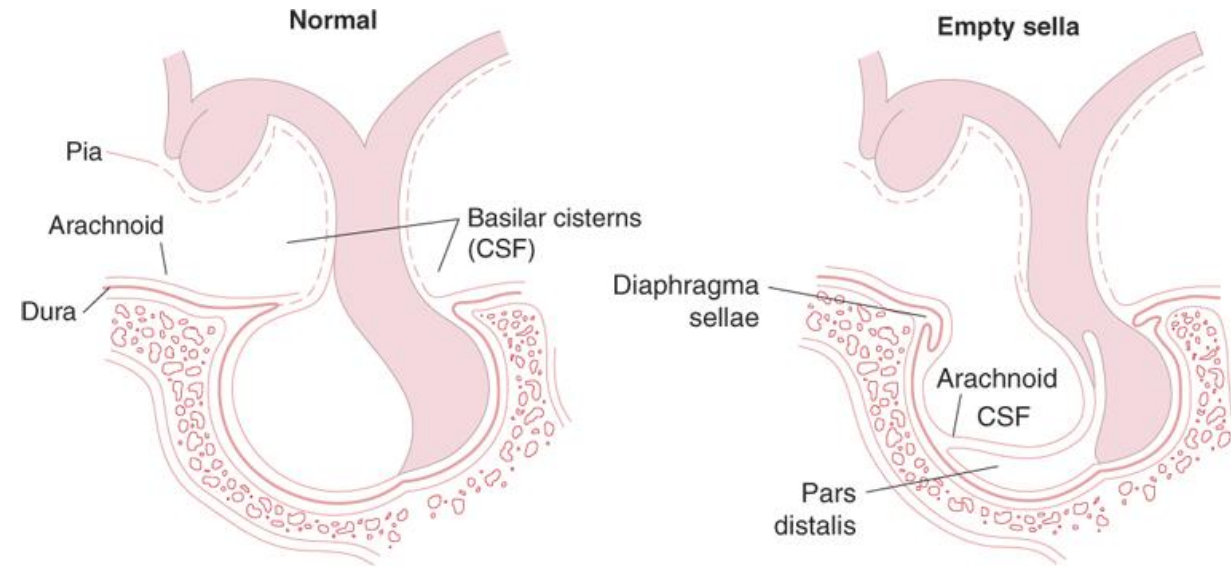
- KAL1 mutation → decreased anosmin-1 expression → impaired migration of GnRH-secreting neurons
- Isolated defect in GnRH secretion + anosmia

Empty sella syndrome

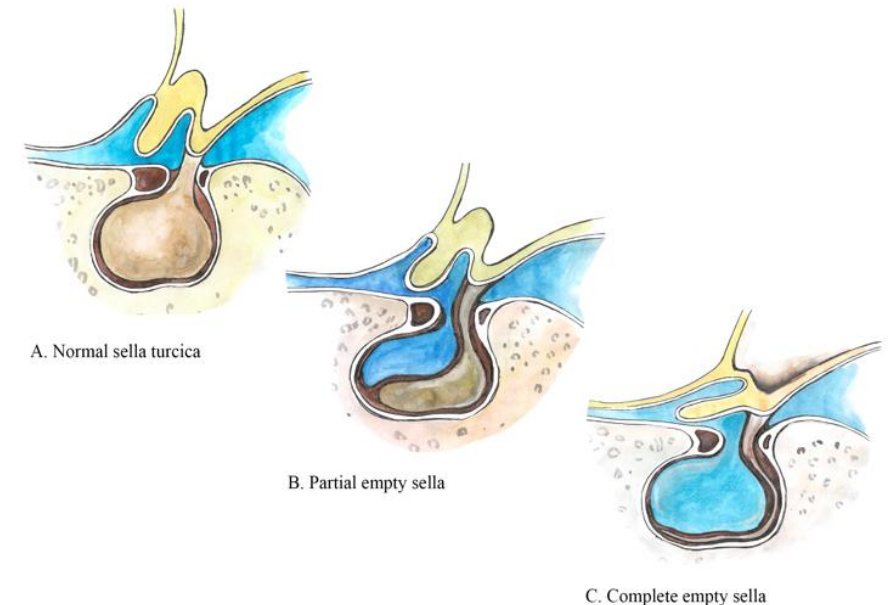
- the subarachnoid space extends into the sella turcica, partially filling it with cerebrospinal fluid → remodeling and enlargement of sella turcica → flattened pituitary

Causes:

- Congenital incompetence of the diaphragma sellae (majority) – obese women with history of multiple pregnancies
- Post pituitary surgery/ radiation therapy
- Postpartum pituitary infraction
- Serious clinical manifestations are uncommon



Source: David G. Gardner, Dolores Shoback: Greenspan's Basic & Clinical Endocrinology, Tenth Edition
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Hypopituitarism

- **Classic course** - initial loss of GH and gonadotropin secretion followed by deficiencies of TSH, then ACTH, and finally PRL.
- Hypogonadism
 - Men- decreased libido, erectile dysfunction
 - Women- secondary amenorrhea
- Prolactin deficiency - failure of postpartum lactation
- Hypothyroidism - Cold intolerance, dry skin, mental dullness, bradycardia, constipation, hoarseness, delayed deep tendon reflexes
- ACTH deficiency - causes adrenocortical insufficiency - Weakness, nausea, vomiting, anorexia, weight loss, fever, and hypotension

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Posterior pituitary disorders

Vasopressin

Decreased vasopressin	Diabetes insipidus
Excess vasopressin	Syndrome of inappropriate anti diuretic hormone secretion (SIADH)

Diabetes insipidus

- excessive urination (polyuria) due to an inability of the kidney to resorb water properly from the urine
- **Etiology:**
 - Central - ADH deficiency
 - Nephrogenic - renal tubular unresponsiveness to circulating ADH
- **Central diabetes insipidus:**
 - Solid tumors – craniopharyngioma, metastatic disease to the pituitary hypothalamic area from lung/ breast cancer
 - Trauma/ surgery at the level of the pituitary/ hypothalamus
 - Granulomatous disease- Sarcoidosis, Wegner granulomatosis, tuberculosis, Langerhan cell histiocytosis

- Cerebral anoxia or brain death
- **Nephrogenic diabetes insipidus:**
 - Genetic - generalized defect in either the V2 class of vasopressin receptors or the aquaporin-2 water channel of the renal collecting ducts.
 - Drug induced – lithium, demeclocycline
 - renal disease that distorts the vascular architecture of the kidney and washes out the inner medulla – polycystic renal disease, renal infarcts, sickle cell nephropathy
 - Hypokalemia, hypercalcemia - decreased expression of aquaporin-2.

Diabetes insipidus

Pathophysiology

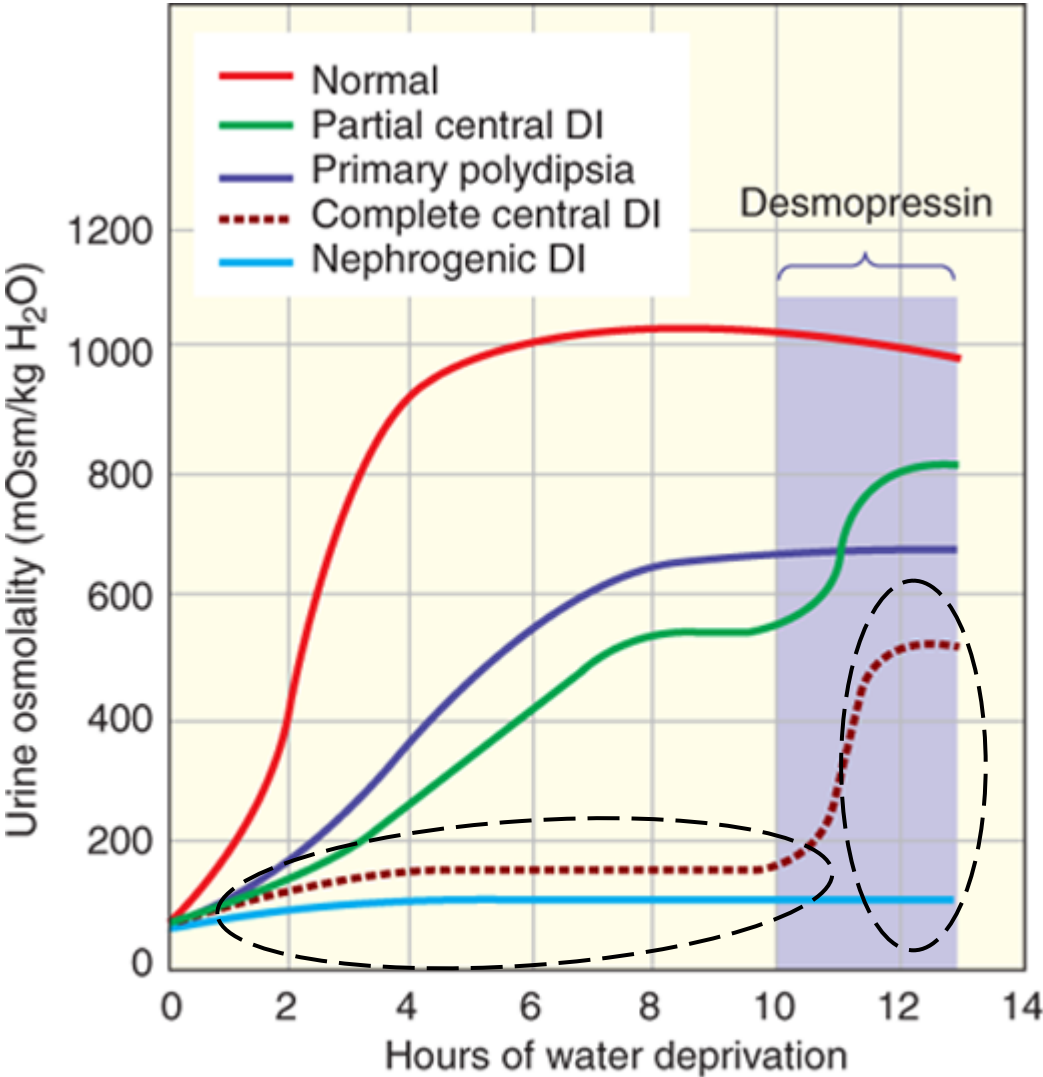
- $UOsm < P_{osm}$ because of excess loss of free water in the urine. ($P_{Osm} > 295 \text{ mOsm/kg}$; $UOsm < 500 \text{ mOsm/kg}$)
- Inability to reabsorb water $\rightarrow$ polyuria
- Increase in plasma osmolality (hypertonicity) $\rightarrow$ polydipsia
- Loss of free water $\rightarrow$ $TBNa^+ / \downarrow \downarrow TBW \rightarrow$ hypernatremia

Euvolemic hypernatremia

Clinical presentation

- Polyuria, nocturia, enuresis
- Dehydration – inability to maintain sufficient water intake
- Hypernatremia - Osmotic forces $\rightarrow$ cell shrinkage and volume loss $\rightarrow$ neurologic manifestations
 - progressive obtundation (decreased responsiveness to verbal and physical stimuli), myoclonus, seizures, focal deficits, and coma

Diabetes insipidus



Water deprivation test

After water restriction	Central diabetes insipidus	Nephrogenic diabetes insipidus
*Urine osmolality	Decreased (<300 mOsm/kg)	decreased (<300 mOsm/kg)

After injection of desmopressin	Central diabetes insipidus	Nephrogenic diabetes insipidus
Urine volume	Decrease	Slight/ no change
*Urine osmolality	Increase (>800 mOsm/kg)	Slight/ no change

*Reference range for urine osmolality = 50-1200mOsmol/kgH₂O

Source: David G. Gardner, Dolores Shoback: Greenspan's Basic & Clinical Endocrinology, Tenth Edition Copyright © McGraw-Hill Education. All rights reserved.

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Syndrome of inappropriate anti-diuretic hormone secretion (SIADH)

Etiology

- Head – **trauma** and CNS disorders – **infection**, inflammation, surgery, **hemorrhage**, mass lesions
- Ectopic production by tumors – **Small cell carcinoma of the lung**
- Lung diseases – pneumonia, tuberculosis, aspergillosis, abscess
- Drug-induced – **Carbamazepine**, cyclophosphamide, SSRIs [**Can't concentrate serum sodium**]

SIADH **HELD**-up water

What do you remember about the action of vasopressin and the consequences of its excess?



Syndrome of inappropriate anti-diuretic hormone secretion (SIADH)

Pathophysiology:

- urine osmolality > plasma osmolality (UOsm > 100 mOsm/kg of water)
- Dilutional hyponatremia due to hypotonic gain of water - Serum Na⁺ < 120 mEq/L; TBNa⁺/↑↑TBW and increase in plasma volume
- Increased urinary sodium > 20 mEq/L

Euvolemic hyponatremia

Clinical presentation:

- Altered mental status – confusion, lethargy, weakness
- Generalized seizures
- Cerebral edema → coma

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Hypothalamic suprasellar tumors

Craniopharyngioma

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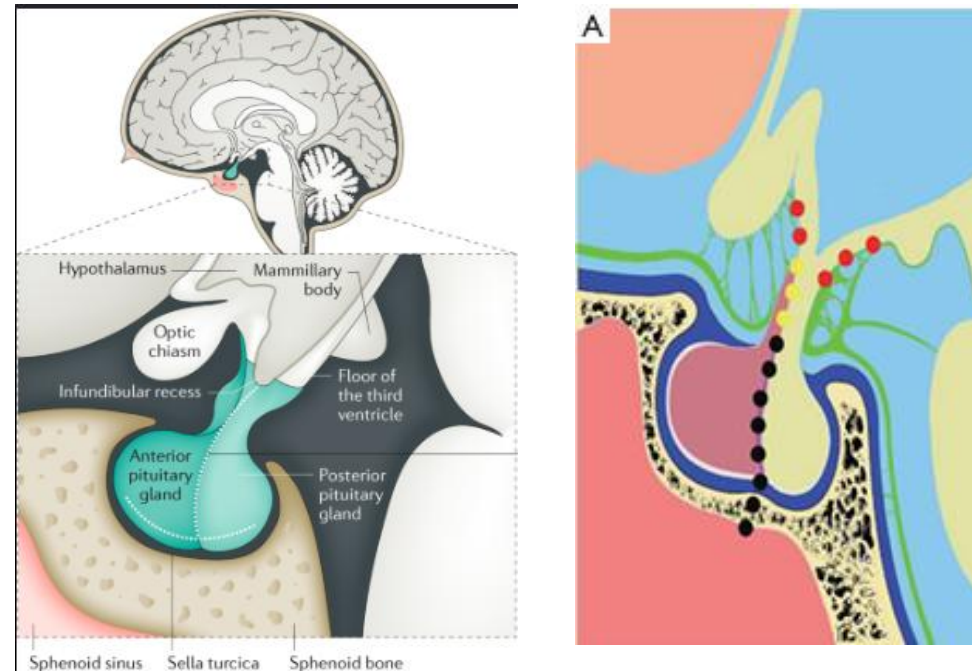
LO6. Recognize craniopharyngioma based on etiopathogenesis, clinical presentation, histopathology and laboratory diagnosis.

Craniopharyngioma

- Benign tumor arising from vestigial remnants of Rathke pouch
- Location- **Suprasellar**. May encroach on the optic chiasm, cranial nerves, floor of the third ventricle and base of the brain.
- Bimodal distribution:
 - **5-15 years** - likely to be adamantinomatous
 - **65 years or older** – likely to be papillary

Clinical presentation:

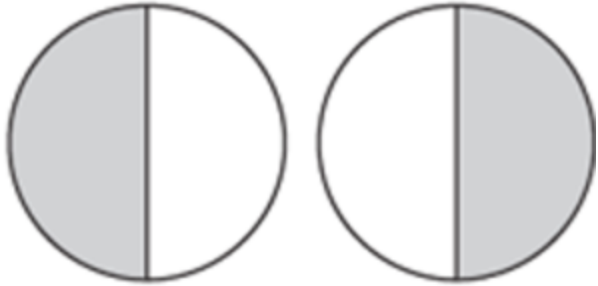
- **Most common cause of hypopituitarism in children**
- Endocrine disturbances precede compression symptoms – **growth retardation** (GH deficiency), arrested puberty (gonadotropin deficiency), **central diabetes insipidus** (pituitary stalk involvement).



Craniopharyngioma

- Neurologic symptoms due to mass effect of the expanding tumor:

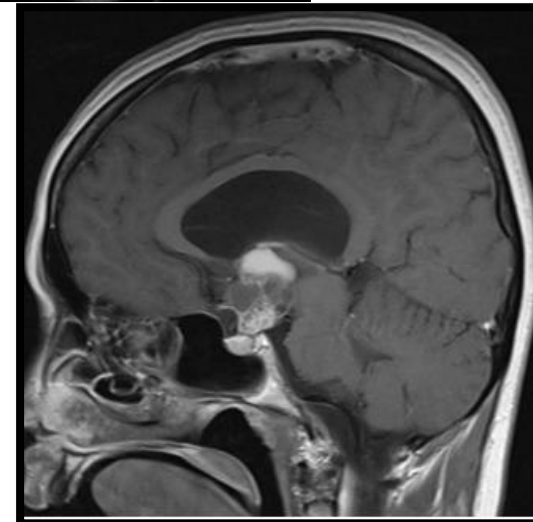
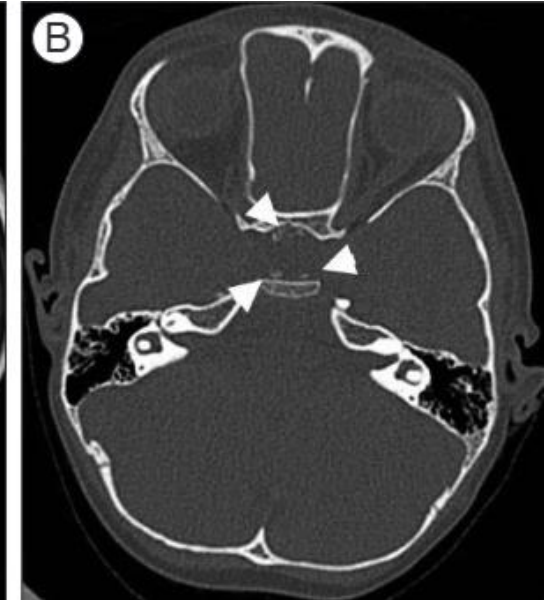
- Compression of optic chiasm
→bitemporal hemianopsia



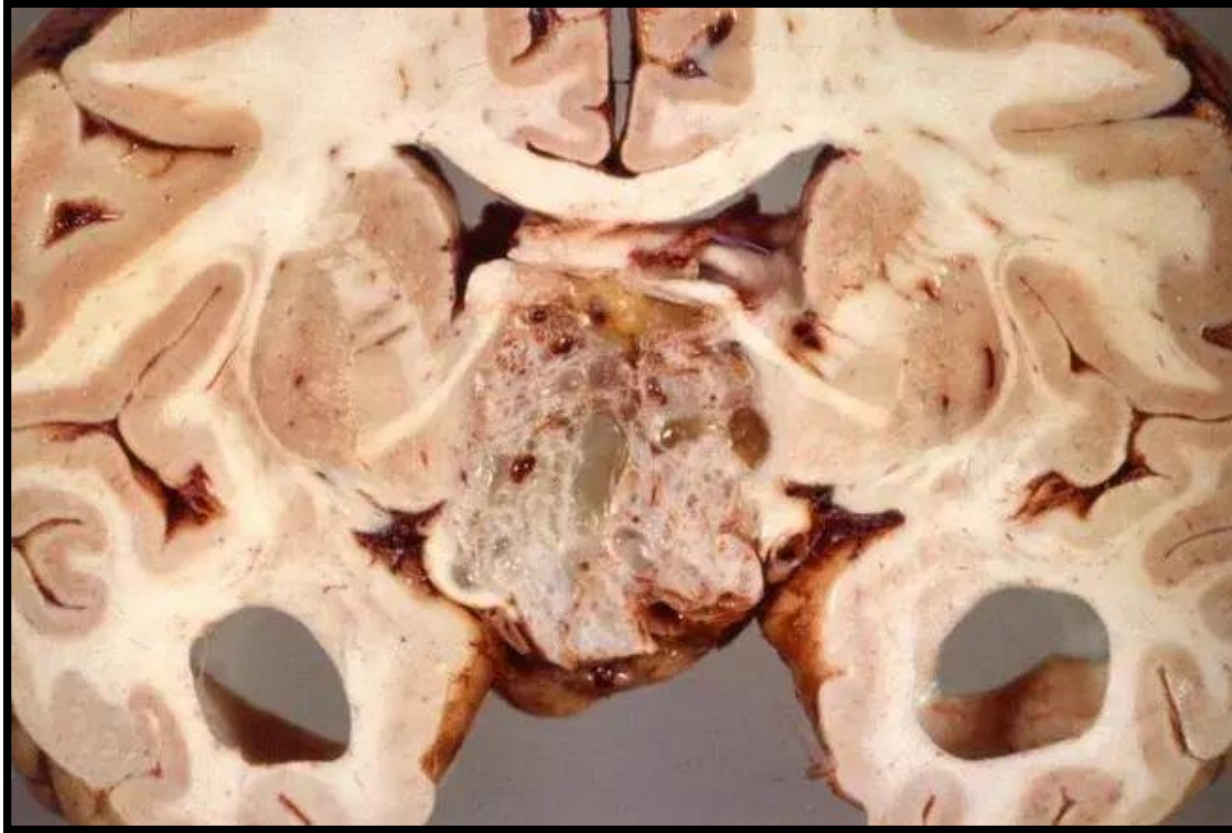
- Increased intracranial pressure due to obstructive hydrocephalus - Headache, vomiting
- Decreased visual acuity, visual field defects

Radiologic findings:

- MRI - heterogeneous, cystic suprasellar mass that enhances with contrast, calcifications may be present.



Craniopharyngioma – Gross morphology

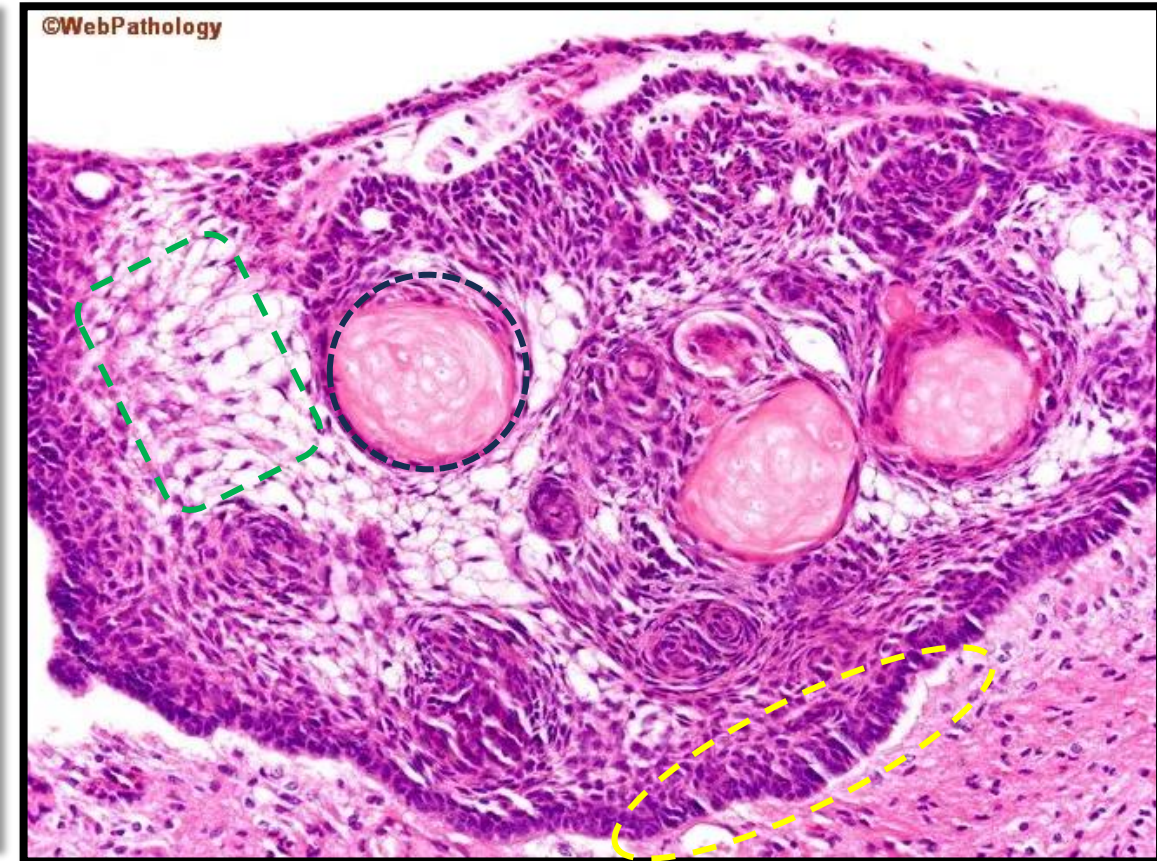
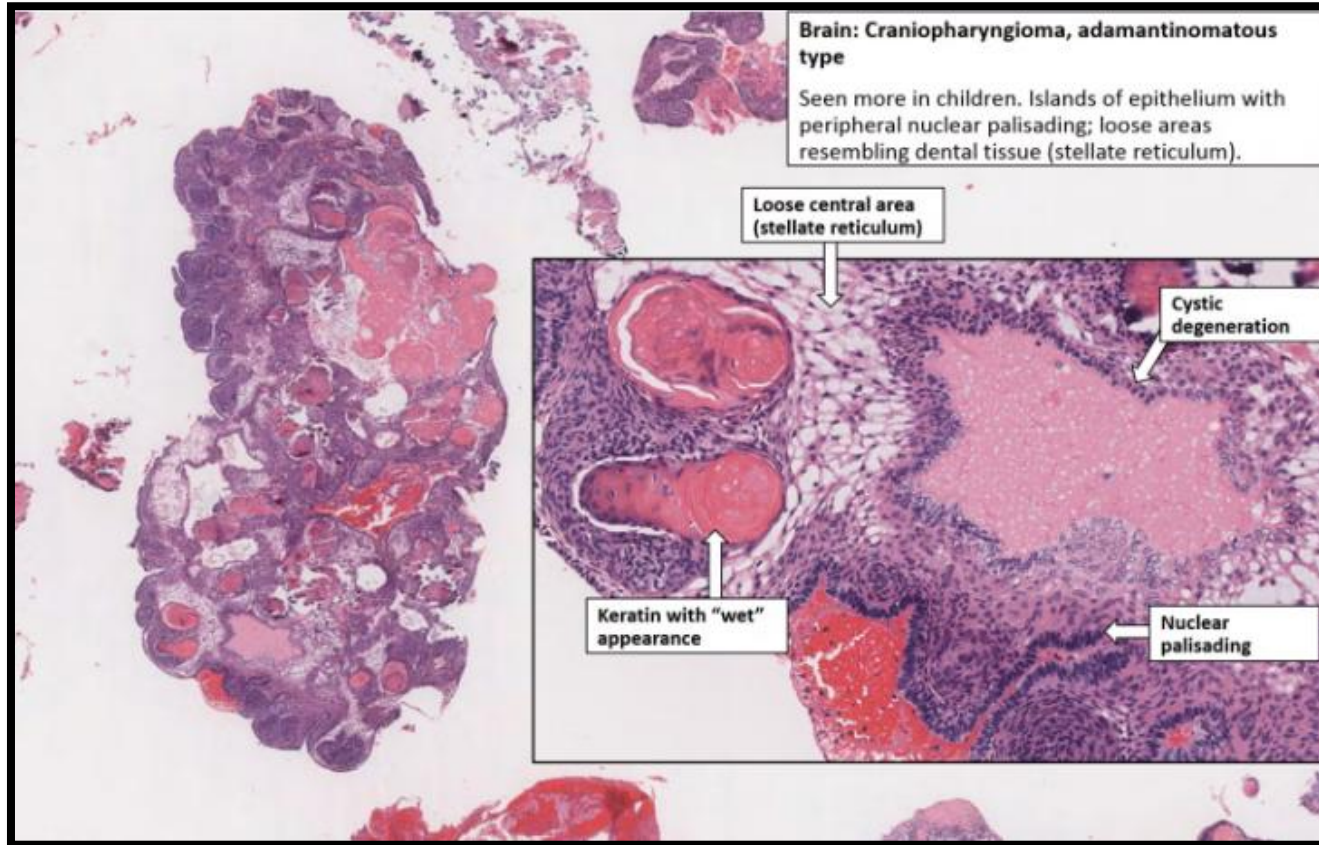


Craniopharyngiomas have **solid and cystic areas** (as seen in this image). **The cystic areas are filled with dark thick fluid rich in proteins, blood, and cholesterol and with an appearance of "motor/machine oil."**



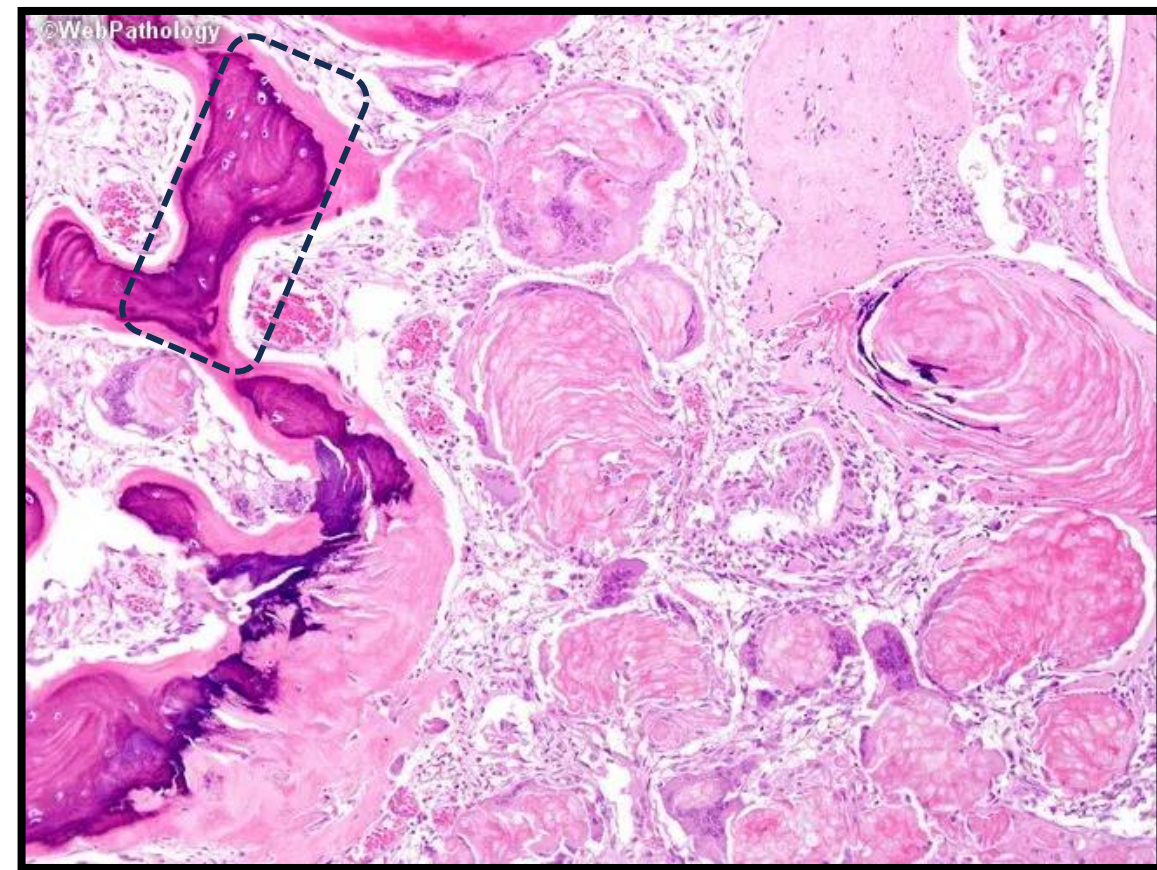
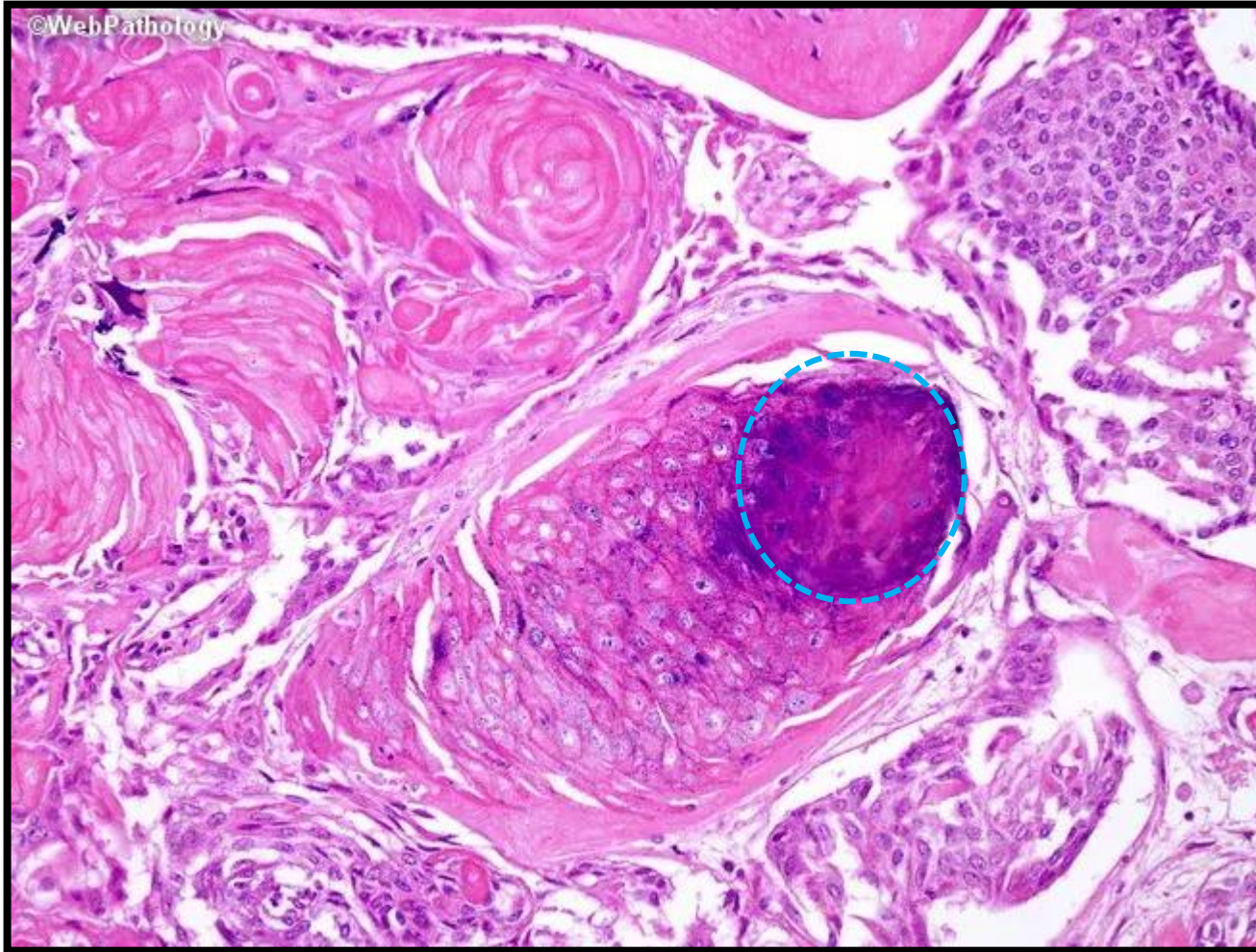
The yellowish-white specks in the right half of the specimen represent the nodules of **"wet keratin"**

Craniopharyngioma - Microscopy



1. Nests or cords of stratified squamous epithelium embedded in a spongy/ **stellate “reticulum”**
2. **“Palisading” of the squamous epithelium** at the periphery
3. Compact, lamellar keratin formation (**“wet keratin”**) - nodules of anucleated squames (ghost cells) with brightly eosinophilic cytoplasm

Craniopharyngioma - Microscopy

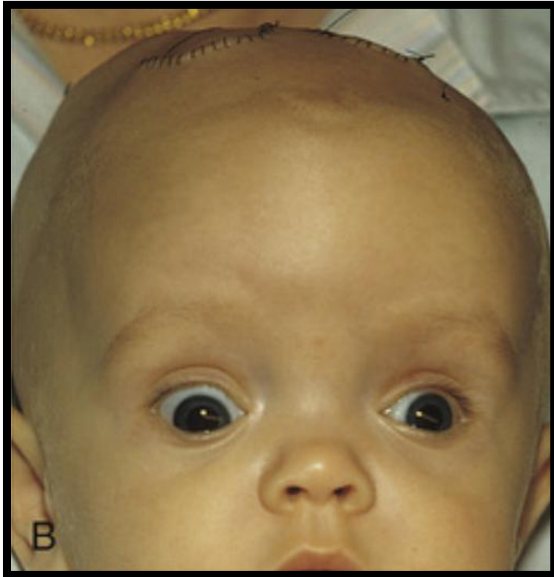


Pineal gland disorders

Pineal gland germinoma

Pineal gland germinoma

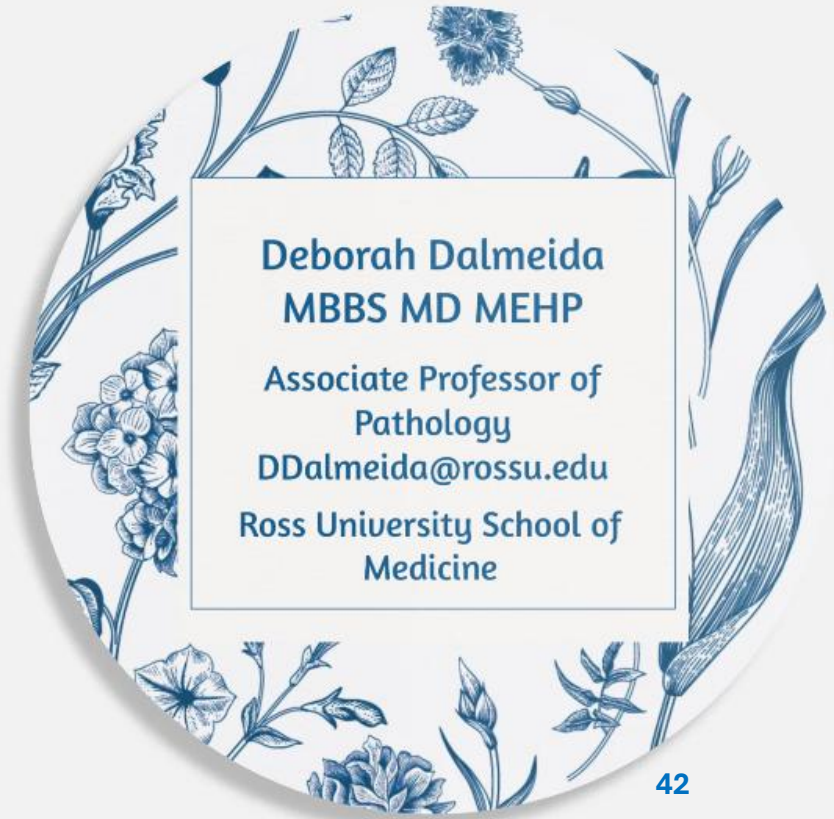
- Most common pineal gland neoplasm and extragonadal germ cell tumor
- Cell of origin- primitive totipotential germ cells
- marked predominance in **males**
- usually occur within the first three decades of life



Clinical presentation:

- **Obstructive (non-communicating) hydrocephalus** due to compression of the cerebral aqueduct draining the third ventricle
- **Parinaud syndrome – triad of:**
 - **upward gaze paralysis** - inability to look up
 - **Light near dissociation** - pupils accommodate to near objects but do not constrict in response to light
 - **Convergence-retraction nystagmus** - eyes jerk back into the orbit on attempted upward gaze

Thank You



Deborah Dalmeida MBBS MD MEHP